## ON-BOARD DIAGNOSTIC SYSTEM PID/DATA MONITOR FUNCTION [TCM (EW6A-EL)]

id0502p7280600

### Purpose/Function

- The PID/DATA monitor function can monitor control data such as input signals/internal calculation values/output signals using the M-MDS.
- The module control condition can be verified in real-time by the PID/DATA monitor function.

### Construction/Operation

### PID/DATA monitor item table

—: Not applicable

| Item | Unit/Condition | Definition |
|---|---|---|
| APP | % | Accelerator pedal position |
| APP1 | % | APP sensor No.1 |
| DGP_DIS_1 | km {mile} | Traveled distance after differential protection control operated |
| DGP_DIS_2 | km {mile} | Traveled distance after DGP_MAX_DIF updated |
| DGP_MAX_DIF | RPM | Maximum rotation difference after differential protection control operated due to excessive rotation difference between left/right drive wheels |
| DGP_SPD | KPH {MPH} | Vehicle speed with trailing wheels after differential protection control operated due to excessive rotation difference between left/right drive wheels |
| ECT | °C {°F} | Engine coolant temperature |
| ECU_A | °C {°F} | ECU internal temperature A |
| ECU_B | °C {°F} | ECU internal temperature B |
| ECU_C | °C {°F} | ECU internal temperature C |
| EOP_DUTY | % | Electric AT oil pump duty cycle |
| EOP_RLY | Off/On | Electric AT oil pump relay condition |
| GEAR_RA | Ratio | Gear ratio |
| GEAR_SEL | 1/2/3/4/5/6 | Gear shift position |
| HI_TEMP | — | ATF high temperature mode determination amount |
| HTM_DIS | km {mile} | Travel distance since determination of ATF high temperature mode |
| I-Stop_EST | Off/On | i-stop enable state on TCM |
| LINEDES | kPa {kgf/cm$^2$, psi} | Target modifier pressure/Target pressure control solenoid pressure |
| LN_C_CLUTCH | kPa {kgf/cm$^2$, psi} | Hydraulic control learning value (engaged) |
| LN_O_CLUTCH | kPa {kgf/cm$^2$, psi} | Hydraulic control learning value (released) |
| LN_OV_SCOPE | — | Hydraulic control learning value (mis-learning) |
| LN_T_CLUTCH | kPa {kgf/cm$^2$, psi} | Hydraulic control learning value (torque converter) |
| LOCK_UP | OFF/SLIP/ON | Torque converter (TCC condition) |
| LONGI_ACCEL | — | Longitudinal acceleration |
| OP_SW1 | Off/On | Oil pressure switch No.1 condition |
| OP_SW1_OFF | kPa {kgf/cm$^2$, psi} | After performing on-board diagnostic test mode |
| OP_SW1_ON | kPa {kgf/cm$^2$, psi} | After performing on-board diagnostic test mode |
| OP_SW2 | Off/On | Oil pressure switch No.2 condition |
| OP_SW2_OFF | kPa {kgf/cm$^2$, psi} | After performing on-board diagnostic test mode |
| OP_SW2_ON | kPa {kgf/cm$^2$, psi} | After performing on-board diagnostic test mode |
| OP_SW3 | Off/On | Oil pressure switch No.3 condition |
| OP_SW3_OFF | kPa {kgf/cm$^2$, psi} | After performing on-board diagnostic test mode |
| OP_SW3_ON | kPa {kgf/cm$^2$, psi} | After performing on-board diagnostic test mode |
| OP_SW4 | Off/On | Oil pressure switch No.4 condition |
| OP_SW4_OFF | kPa {kgf/cm$^2$, psi} | After performing on-board diagnostic test mode |
| OP_SW4_ON | kPa {kgf/cm$^2$, psi} | After performing on-board diagnostic test mode |
| OSS | RPM | Output shaft speed |
| PADDLE_DOWN*1 | Off/On | Down switch condition (steering shift switch) |
| PADDLE_UP*1 | Off/On | Up switch condition (steering shift switch) |
| PUMP_SPEED | RPM | Motor pump unit actual speed |
| RPM | RPM | Engine speed/min |
| SC_STATE | Not Active/Active | Shift control execution condition |

| Item | Unit/Condition | Definition |
|---|---|---|
| SE_TYPE | No valid data/Bf_1st<br>Af_2nd/Bf_1st<br>Af_3rd/Bf_1st<br>Af_4th/Bf_1st<br>Af_5th/Bf_1st<br>Af_6th/Bf_2nd<br>Af_1st/Bf_2nd<br>Af_3rd/Bf_2nd<br>Af_4th/Bf_2nd<br>Af_5th/Bf_2nd<br>Af_6th/Bf_3rd<br>Af_1st/Bf_3rd<br>Af_2nd/Bf_3rd<br>Af_4th/Bf_3rd<br>Af_5th/Bf_3rd<br>Af_6th/Bf_4th<br>Af_1st/Bf_4th<br>Af_2nd/Bf_4th<br>Af_3rd/Bf_4th<br>Af_5th/Bf_4th<br>Af_6th/Bf_5th<br>Af_1st/Bf_5th<br>Af_2nd/Bf_5th<br>Af_3rd/Bf_5th<br>Af_4th/Bf_5th<br>Af_6th/Bf_6th<br>Af_1st/Bf_6th<br>Af_2nd/Bf_6th<br>Af_3rd/Bf_6th<br>Af_4th/Bf_6th<br>Af_5th | Gear shift position before shifting gears |
| SHIFT_CTRL | DEFAULT/<br>MANUAL/<br>C_CONTROL/<br>HIGH_TEMP/<br>D_MANUAL/<br>FAIL_SAFE | Shift control mode |
| SS_ON-OFF | Off/On | On/off solenoid condition |
| SS1 | A | Shift solenoid No.1 actual current |
| SS1_C | A | Shift solenoid No.1 target current |
| SS2 | A | Shift solenoid No.2 actual current |
| SS2_C | A | Shift solenoid No.2 target current |
| SS3 | A | Shift solenoid No.3 actual current |
| SS3_C | A | Shift solenoid No.3 target current |
| SS4 | A | Shift solenoid No.4 actual current |
| SS4_C | A | Shift solenoid No.4 target current |
| SSLU | A | TCC control solenoid actual current |
| SSLU_C | A | TCC control solenoid target current |
| SSLU_PRES | kPa {kgf/cm$^2$, psi} | TCC control solenoid target pressure |
| SSP | A | Pressure control solenoid actual current |
| SSP_C | A | Pressure control solenoid target current |
| SSP_PRES | kPa {kgf/cm$^2$, psi} | Pressure control solenoid target pressure |
| TFT | °C {°F} | ATF temperature |
| THOP | % | Throttle opening angle |
| TORQUE_ACT | Nm | Actual engine torque |
| TORQUE_DES | Nm | Desired engine torque |
| TR | P/SHIFT_DOWN/<br>SHIFT_UP/M/D/N/<br>R | Range position |
| TS_OD | km {mile} | Traveled distance when AT slip occurred |

| Item | Unit/Condition | Definition |
|---|---|---|
| TS_SE | No valid data/Bf_1st<br>Af_2nd/Bf_1st<br>Af_3rd/Bf_1st<br>Af_4th/Bf_1st<br>Af_5th/Bf_1st<br>Af_6th/Bf_2nd<br>Af_1st/Bf_2nd<br>Af_3rd/Bf_2nd<br>Af_4th/Bf_2nd<br>Af_5th/Bf_2nd<br>Af_6th/Bf_3rd<br>Af_1st/Bf_3rd<br>Af_2nd/Bf_3rd<br>Af_4th/Bf_3rd<br>Af_5th/Bf_3rd<br>Af_6th/Bf_4th<br>Af_1st/Bf_4th<br>Af_2nd/Bf_4th<br>Af_3rd/Bf_4th<br>Af_5th/Bf_4th<br>Af_6th/Bf_5th<br>Af_1st/Bf_5th<br>Af_2nd/Bf_5th<br>Af_3rd/Bf_5th<br>Af_4th/Bf_5th<br>Af_6th/Bf_6th<br>Af_1st/Bf_6th<br>Af_2nd/Bf_6th<br>Af_3rd/Bf_6th<br>Af_4th/Bf_6th<br>Af_5th | Gear position before/after shifting when AT slip occurred |
| TS_VS | KPH {MPH} | Vehicle speed before/after shifting when AT slip occurred |
| TSS | RPM | Turbine/input shaft speed |
| UPSHIFT_REV | Off/On | Shift-up record due to engine request |
| VPWR | V | TCM power supply voltage |
| VSS | KPH {MPH} | Vehicle speed |

*1 : With steering shift switch